

Phase 1 Report

BMP.tn – Digital Marketplace & Management Platform

for the Construction Sector

Executive Summary

BMP.tn is a responsive digital platform connecting artisans, prescribers (architects/engineers), suppliers (manufacturers), and administrators. The platform combines marketplace features with an artisan-adapted ERP workflow (projects, quotes, invoices, orders) to improve efficiency, cost control, and adoption in the construction sector.

1. Business Features Specification

1.1 Artisan

- Create and manage projects (chantier).
- Generate quotes (devis) and invoices (factures).
- Create and track orders to suppliers.
- Track costs and margins to understand profitability.
- Access product catalogs and technical documentation.

1.2 Prescriber (Architect / Engineer / Interior Architect)

- Browse products and solutions by category.
- Access technical documentation (datasheets, manuals, videos).
- Search artisans by region and specialty and contact them.

1.3 Supplier (Manufacturer / Vendor)

- Manage product catalog (create/update products).
- Attach documentation and media to products.
- Increase visibility to prescribers and artisans.

1.4 Administrator

- Manage users and roles (RBAC).
- Moderate content quality and handle abuse reports.
- Manage subscriptions (abonnement) and monitor payments status.

- Access platform analytics dashboards.

2. Product Backlog (User Stories)

2.1 Artisan User Stories

- As an artisan, I want to create a project so that I can manage my chantier digitally.
- As an artisan, I want to generate a quote so that I avoid manual calculation errors.
- As an artisan, I want to generate an invoice so that I can bill my client quickly.
- As an artisan, I want to track expenses per project so that I know my profit margin.
- As an artisan, I want to place an order to a supplier so that I save time and reduce stock issues.
- As an artisan, I want to subscribe to a plan so that I can access advanced ERP features.

2.2 Prescriber User Stories

- As a prescriber, I want to browse products by solution type so that I can choose suitable materials.
- As a prescriber, I want to find artisans by region and specialty so that I can build a project team.

2.3 Supplier User Stories

- As a supplier, I want to add products to my catalog so that users can discover my solutions.
- As a supplier, I want to update documentation so that product information stays accurate.

2.4 Admin User Stories

- As an admin, I want to manage roles and permissions so that data access stays secure.
- As an admin, I want to monitor subscriptions so that access is correctly enabled/disabled.

3. Technical Requirements

Technology stack selected for the MVP: React (frontend), Node.js + Express (backend), MongoDB (database).

3.1 Frontend (React)

- Responsive UI (mobile-first) for artisans working on construction sites.
- Role-based navigation (Artisan, Prescriber, Supplier, Admin).
- Forms and dashboards for projects, products, quotes, invoices, orders.

3.2 Backend (Node.js + Express)

- REST API exposing core resources (users, products, projects, quotes, invoices, orders, subscriptions).
- Authentication and authorization (JWT recommended).

- Role-Based Access Control (RBAC) and input validation.

3.3 Database (MongoDB)

- Document-based storage for flexible MVP evolution.
- Indexes on key fields (email, role, supplierId, artisanId, projectId).
- Data separation by ownership (each artisan accesses only their own business data).

4. Architecture of the Application

The platform follows a three-tier architecture:

- Client Layer (React): UI and user interactions.
- Server Layer (Node.js/Express): business logic and REST APIs.
- Data Layer (MongoDB): persistent storage.

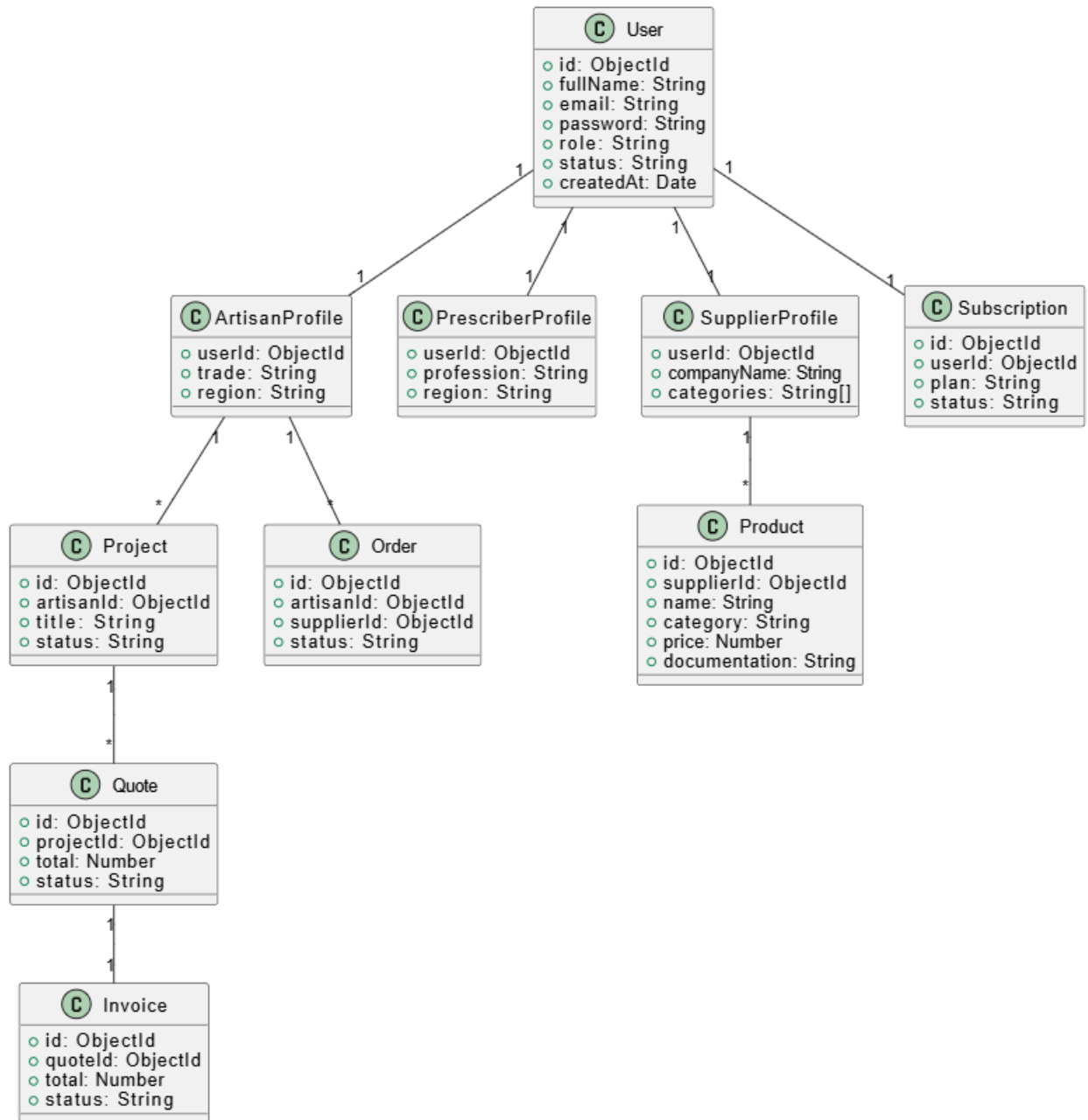
Optional integrations (future): payment gateway, notifications (email/SMS).

5. Data Models

MongoDB collections are structured around a base User plus role-specific profiles. Business objects include Project, Quote, Invoice, Product, Order, and Subscription.

5.1 Class Diagram (Conceptual Data Model)

Figure 1 – Class diagram for BMP.tn (User, Profiles, Business Entities).



6. Automation and Continuous Integration

- GitHub used for version control and team collaboration.
- GitHub Projects used for backlog, sprints, and milestones.
- GitHub Actions used for CI: install dependencies, run lint, run tests (when available), and build validation.

7. Prototype

A UI prototype (Figma or mockups) is produced for Phase 1 to validate main user journeys. Recommended screens: login/signup, artisan dashboard, project creation, quote generation, product browsing.

8. Conclusion

This Version report defines the business scope, backlog, technical choices, architecture, and conceptual data model of BMP.tn. The platform is designed as a responsive solution that prioritizes artisan adoption while connecting prescribers and suppliers in a unified ecosystem.